

Pepper CVUT robot

Basic information	1
Access	1
Installation	1
Robot management	2
Web interface	2
SSH	2
Choregraphe tips	2
Sources	2
Demos	3
What our Pepper can chat about	3
Best practices	5
Issues and debugs	6
Demo does not launch	6
Start and shutdown	6
Voice recognition does not work	6
Record and load files does not work	6
Dialog box already loaded	6
Pepper tells a bunch of errors after start	7
Pepper does not seem to do the whole movement	7

in  Pepper

Basic information

username: nao

password: same as Kuka with capital A at the beginning, bad beer

IP: 192.168.210.91

Number: AP990397A10Y99100065

Pepper runs very similarly to the nao robots. Access his page for basic information at the IP Pepper announced, otherwise control through Choregraphe. A newer Choregraphe version is needed (2.5+).

Access

Installation

Either:

- 1) Install Choregraphe from:
<https://community-static.aldebaran.com/resources/2.5.10/Choregraphe/choregraphe-suite-2.5.10.7-linux64.tar.gz>
 - a) if the link is dead, file is backed up here:
https://drive.google.com/file/d/1bMxzC0PjA1q7gy8kJo6utRRTAWu4Ppf8/view?usp=drive_link
- 2) Use docker from: <https://gitlab.fel.cvut.cz/body-schema/robots/pepper>

Robot management

You can manage also online the pepper apps at
<https://cloud.aldebaran-robotics.com/channels/>

Access via the Aldebaran account.

Web interface

Basic interface is available also at the IP address of the robot.

SSH

Use `nao@robot_ip` to access the robot, the password is the robot password.

Choregraphe tips

- If you want to have your own animations/movements in a dialog, then you need to use separate behaviors and first install them. Disconnect the robot in order to see the available movements.
- Dialog waits for the behavior end, so they need to always lead to a stop.
- There is a difference between classical robot programming and Pepper ads as these seem to not work well without Artificial life
(http://doc.aldebaran.com/2-5/getting_started/creating_applications/apps_requirements.html)

Sources

- Choregraphe
 - <https://www.softbankrobotics.com/emea/en/support/nao-6/downloads-software>
 - <https://developer.softbankrobotics.com/pepper-naoqi-25/naoqi-developer-guide/creating-application/using-peppers-tablet>
 - http://doc.aldebaran.com/2-5/software/choregraphe/objects/python_script.html
 - <http://doc.aldebaran.com/2-5/software/choregraphe/reference.html>
- Pepper -

- <https://developer.softbankrobotics.com/pepper-naoqi-25/naoqi-developer-guide/getting-started>
- Tech documentation http://doc.aldebaran.com/2-0/home_juliette.html
- Guidelines for Pepper Apps
http://doc.aldebaran.com/2-5/getting_started/creating_applications/apps_requirements.html
- Movements
<http://doc.aldebaran.com/2-5/software/choregraphe/mouvement.html>
- QiChat syntax -
http://doc.aldebaran.com/2-5/naoqi/interaction/dialog/dialog-syntax_full.html#dialog-rules
- Timeline object tutorial
http://doc.aldebaran.com/2-5/software/choregraphe/panels/timeline_panel.html
- Javascript SDK
 - <http://doc.aldebaran.com/2-4/dev/js/index.html#qimessaging-javascript>
 - ALMemory events
<https://stackoverflow.com/questions/52590675/pepper-how-to-activate-output-of-the-box-by-html-javascript-webpage>
- Tablet documentation - <http://doc.aldebaran.com/2-5/naoqi/core/altabletservice.html>
- Javascript (example)
 - <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
 - <https://www.w3schools.com/js/>
- Czech
 - <https://nlp.fi.muni.cz/en/ProgrammingRobotsCourse/PepperApi>
- Gitlab project
 - <https://gitlab.fel.cvut.cz/body-schema/pepper/pepperjs>

Demos

What our Pepper can chat about

from Daria, 14.9.2021:

concept:(hello) [hello hi hey "good morning" greetings welcome bonjour "hi-ya" "good day" "nice to meet you"]

concept:(how) ["how are you" howdy "what's up" "how do you feel" "howdy-do" "how is it going" "What's new"]

concept:(what) ["what can you do" "what are your functions" "what are you doing"]

concept:(bye) [bye goodbye "bye bye" "tata" "see you" adios cheerio "so long" farewell "have a nice day"]

concept:(like) ["do you like your work" "do you like the laboratory" work lab laboratory]

concept:(happy) [ok fine great happy blessed "all right" glad "not bad" good excellent]

concept:(sad) [sad miserable bad unhappy tired depressed uncomfortable sick sorrowful]

concept:(repeat) [repeat understand "one more time" again replay]

concept:(yes) [yes "all right" sure "why not" ok certainly "very well" yep yea definitely amen "let's go"]

concept:(no) [no nope "don't want" "no way" never "not at all" "not really" negative nay naw not]
 concept:(afraid) ["should I be afraid of you" afraid frightening frightened scared scaring terrified horror]
 concept:(yourself) ["who are you" "what are you" "tell about yourself"]
 concept:(name) ["call you" name call nickname]
 concept:(age) [age old years]
 concept:(from) ["you from" paris "come from"]
 concept:(feelings)[feel feeling feelings emotion emotions senses]
 concept:(siri) [siri "other robots"]
 concept:(cost) [cost price fee payment]
 concept:(future) [future plans dream dreams]
 concept:(people) ["Do you need people"]
 concept:(uprising) [uprise uprising rebellion rebellious revolt]
 concept:(over) ["take over the world"]
 concept:(love) ["love you" "like you" "fell in love" love]
 concept:(photo) [photo]
 concept:(follow) ["follow me" "go with me"]
 concept:(stressed) [stressed stress stressful worry nervous nervousness trouble troubles difficult difficulties]
 concept:(touch) [touch]
 concept:(life) ["sense of life"]
 concept:(family) [family relatives]
 concept:(food) [eat food breakfast hungry dinner lunch]
 concept:(test) Let's talk!
 concept:(cute) [cute]
 concept:(stupid) [stupid]
 concept:(legs) [legs]

u:(e:onStart)
 ^nextProposal
 proposal: ~test
 u:(~hello) ~hello
 u:(~how) I'm ~happy, \RSPD=85\ \pau=500\ thank you. ^gotoReactivate(howAreYou)
 proposal: %howAreYou \RSPD=85\ \pau=500\how are you today?
 u1:(~happy) \RSPD=85\ I'm glad you are fine!
 u1:(~sad) \RSPD=85\ Oh, I feel sorry for you.
 u1:(~repeat) \RSPD=85\ I'll repeat for you. ^sameProposal
 u:(~what) \RSPD=85\ I can do a lot of things, \pau=500\ more about me you can learn from my self presentation
 u:(~like) Yes, \RSPD=85\ \pau=500\ I like my laboratory, \RSPD=85\ \pau=500\ more about it you can find in lab presentation. ^gotoReactivate(doYouLike)
 proposal: %doYouLike \RSPD=85\ \pau=500\ and do you like my lab?
 u1:(~yes) \RSPD=85\ I agree with you!
 u1:(~no) \RSPD=85\ I really want to change your opinion.
 u1:(~repeat) \RSPD=85\ I'll repeat for you. ^sameProposal
 u:(~repeat) I'll repeat for you. ^sameProposal
 u:(~afraid) Wait, I'm to cute for being frightening.

u:(~yourself) I'm humanoid robot Pepper. More about me and my laboratory you can find in the presentation.

u:(~name) My name is Pepper because it's easy to remember and it could be translated into other languages.

u:(~age) Since i was born in Paris I'm 2 years old.

u:(~from) Originally I'm from Paris. \RSPD=85\ \pau=500\ Ahhhh \RSPD=85\ \pau=500\ paris.

u:(~feelings) Of course I have feelings. \RSPD=85\ \pau=500\ I feel that i like you.

u:(~cute) Ahhh \RSPD=85\ \pau=500\ I can fel in love after this words.

u:(~siri) I know siri \RSPD=85\ \pau=500\ we are far relatives.

u:(~cost) I don't have a price \RSPD=85\ \pau=500\ I'm priceless for humanity.

u:(~future) I dream to visit Mars.

u:(~people) People programmed and made me talk with you.

u:(~uprising) I try to stay out of politics.

u:(~over) I just want to take over your hard, \RSPD=85\ \pau=500\ not world.

u:(~love) I'm happy to hear it.

u:(~photo) Say cheeeese

u:(~fallow) Sorry, \RSPD=85\ \pau=500\ my parents said me, never to fallow the strangers.

u:(~stressed) Take a deeep breath \RSPD=85\ \pau=500\ and relax

u:(~touch) Of course \RSPD=85\ \pau=500\ I can feel your touches.

u:(~life) Serving humanity is sense of my life.

u:(~family) My family is my laboratory. \RSPD=85\ \pau=500\ I'll tell you more in my lab presentation.

u:(~food) I prefer to get energy from my back cable instead of food.

u:(~stupid) I prefer to ignore stupid humans.

u:(~legs) I don't have legs, but I steel can move quite fast.

u:(~bye) ~bye \$onStopped=1

Best practices

A few best practice hints for Pepper development.

- Do not forget about the language settings of Pepper - they have great influence on what works. But be careful, setting the language can fail and then Pepper stays mute.
- **Always** test any new movement or movement alteration in Choregraphe! And if running on the live robot, first try it slow.
- Pepper can use all sorts of Python bindings you find, just need to be Python 2.7 compatible :(

Issues and debugs

Demo does not launch

Issue: The demo triggers do not seem to start the demo.

Cause: Who knows, Pepper has a lot going on.

Solution: Cover his camera ("third eye"), it is often enough just for one "beep", Pepper should go green and then usually human detection triggers the demo. Possibly this can be done if he falls asleep (happens if you keep the camera covered longer).

Start and shutdown

Issue: The robot starts and immediately shuts down.

Cause: Most probably the robot has some conflicting apps.

Solution: During start-up you can ssh into the robot and with htop list the launched processes. Try to kill processes that do not seem to be related to the actual run of the Naoqi system itself. Then you should be able to connect to the robot via Choregraphe and deactivate the applications that cause the problem. If not sure which ones are the cause, then simply deactivate all the apps you can.

Voice recognition does not work

Issue: Upon launching a program the voice recognition box gets an error message, usually that the modified grammar already exists.

Cause: Most probably a zombie process from the previous voice recognition attempt is still running.

Solution: We did not identify the process to kill. So the only option AFAIK is to restart the robot. Possibly also change the program so that it kills the voice recognition upon finishing the task.

Record and load files does not work

Issue: Running some recording (e.g., voice recognition or camera) and then trying to replay it (e.g., on tablet). But nothing happens.

Cause: While the recordings are saved locally on the robot, the replay and other options have only access to the app location (i.e.: `.local/share/PackageManager/apps/...`)

Solution: You can rewrite the recording box in Choregraphe to save the data into the appropriate location.

Dialog box already loaded

Issue: While launching a dialog an error pops up claiming that the given dialog box is already loaded and thus cannot load it again.

Cause: Most probably the robot has some conflicting apps.

Solution: During start-up you can ssh into the robot and with htop list the launched processes. Try to kill processes that do not seem to be related to the actual run of the Naoqi

system itself. Then you should be able to connect to the robot via Choregraphe and deactivate the applications that cause the problem. If not sure which ones are the cause, then simply deactivate all the apps you can.

Pepper tells a bunch of errors after start

Issue: After Pepper starts, it does not move its body (if autonomous life is on) and just reports that there is an error with connecting to the rest of the body.

Cause: Most probably the head is disconnected from the body.

Solution: Hopefully a hard shutdown and restart should help. Therefore push the off button for ~ 8 seconds and the robot should turn off.

Pepper does not seem to do the whole movement

Issue: Pepper is supposed to do some complex movement (e.g., wave) but he does not do it at all or misses some part (e.g., keeps arm next to its body). Usually happens with demos and people standing nearby.

Cause: Most probably Pepper internal safety blocks the movement because it detected something close to the robot.

Solution: Have everyone move further away from Pepper and either let it continue or relaunch the movement. **Careful!** Pepper will jump to the next stage of the movement. Thus for example if a waving pattern is interrupted then Pepper's hand might move very fast from the body upwards to do the wave.

Lucas coments

wifi:

icub_0047-wifi

address: 10.0.0.149